

1.) A Shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them. A wolf approaches the flock and calls out, "Wolf!" the villagers refuse to be fooled. The town goes hungry. Panic ensues.

Act \ pre		
	Y	N
Y	TP	FN
N	FP	TN

- TP - Real wolf came, warning also given
- FP - No wolf false warning given
- FN - No wolf will come, wolf came, predicted it won't come.
- TN - No wolf, no false alarm

2.) Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat image in 50 of 70 cat images. It also predicts absence of a cat in 20 of 70.

Act \ pre	Yes	
	CP	NCP
Cat (30)	25	20
NO Cat (70)	5	50

• Total image = 100 and 70.

• Actual cat images = 30

• Actual no-cat images = 70

precision : $\frac{TP}{TP + FP} = \frac{25}{25 + 5} = 0.833$